Pankaj Sharma(ps38984)

Project Plan: Translation Confidence Scoring

(Predicting When Machine Translation Will Fail)

Course: Fall26 - CASE STUDIES/MCHNE LEARNING-WB (54530) **Date: 10/12/2026**

1. Topic Summary

Machine translation (MT) systems such as Google Translate or NLLB are fluent but not always correct. They can silently drop information, mistranslate terms, or produce "hallucinated" output, and a reader without knowledge of the source language cannot tell. This project builds a model that predicts how reliable a given translation is **without needing a human reference translation**. This task is known as Machine Translation Quality Estimation (QE).

Given a source sentence and its machine translation, the system will output either a confidence score (for example, 0 to 100) or a label ("likely OK" or "likely to fail"). Such a system is useful in practice: translations with low confidence can be routed to human reviewers, while high-confidence ones can be published automatically, saving time and cost.

The project will focus on **one language pair** (English to Hindi) to keep the scope manageable and allow a thorough analysis.

2. Research Questions

1. How accurately can a model predict translation quality using only the source sentence and its machine translation?
2. Do classical machine learning models with hand-crafted features (Random Forest, XGBoost) perform comparably to a fine-tuned XLM-RoBERTa-base model?
3. Which features or signals are most informative of translation failure (sentence length ratio, language model perplexity, embedding similarity between source and translation)?
4. What types of sentences cause the most failures (long sentences, idioms, rare words, named entities, numbers)?
5. Can the predicted score be turned into a reliable decision threshold for "send to human review" with acceptable precision and recall?

3. Data, Models and Approach

Data

- **Primary dataset:** WMT Quality Estimation shared task data or MLQE-PE, which contain source sentences, machine translations and human quality scores (direct assessment scores or post-editing effort, HTER) for several language pairs, including English-Hindi.
- **Backup option:** Generate translations for a public parallel corpus (e.g., Multi30k or a subset of Tatoeba) using a pre-trained MT model, then derive quality labels by comparing against reference translations with metrics such as chrF or COMET.

Pipeline

1. **Data preprocessing:** clean text, remove duplicates and empty or misaligned pairs, normalize Unicode, tokenize, and normalize quality scores. Split the data into training, validation and test sets.
2. **Feature engineering (classical approach):** source and target length, length ratio, token and punctuation overlap, number and named-entity consistency, language model perplexity, and cosine similarity between multilingual sentence embeddings (LaBSE or multilingual BERT).
3. **Model training:**
    - *Baselines:* Linear Regression, Random Forest, XGBoost on engineered features.
    - *Advanced:* fine-tune XLM-RoBERTa-base only, with a regression head on concatenated source and translation pairs.
4. **Prediction:** output a continuous confidence score, plus a binary "fail" label using a tuned threshold.
5. **Evaluation:**
    - Regression: Pearson and Spearman correlation, RMSE, MAE.
    - Classification: precision, recall, F1, ROC-AUC.
    - Error analysis on failure cases and feature importance (e.g., SHAP).

Expected Outcome

A comparison table of the baseline models versus the fine-tuned XLM-RoBERTa-base model, an analysis of what drives translation failure, and a working prototype that flags low-confidence translations.

Tools

Python, pandas, scikit-learn, XGBoost, Hugging Face Transformers, PyTorch, Matplotlib/Seaborn.

4. Optional Extension: English-Spanish

If the main English-Hindi work is finished with time to spare, the same pipeline can be repeated on English-Spanish. Hindi is a lower-resource language and Spanish a high-resource one, so this adds a comparison of whether translation failures are easier to predict in one than the other.

- **Data:** check whether WMT Quality Estimation data with quality labels exists for English-Spanish; if not, use the backup route (translate a public parallel corpus and derive labels with chrF or COMET).
- **Approach:** reuse the same preprocessing, features, baselines and XLM-RoBERTa-base fine-tuning, then compare results across the two language pairs.
- **Scope:** this section is a stretch goal and not required for the core project.
- **Optional demo:** a small Streamlit app where a user enters a sentence, receives a translation and sees its confidence score.

5. Questions for Guidance

1. Is focusing on a single language pair sufficient in scope, or should a second pair be included for comparison?
2. Would you prefer the task framed as regression (predicting a score) or classification (OK vs. fail), or is covering both acceptable?
3. Is using the pre-trained XLM-RoBERTa-base transformer appropriate for this project?
4. Are there recommended evaluation standards or baselines expected for this kind of project?